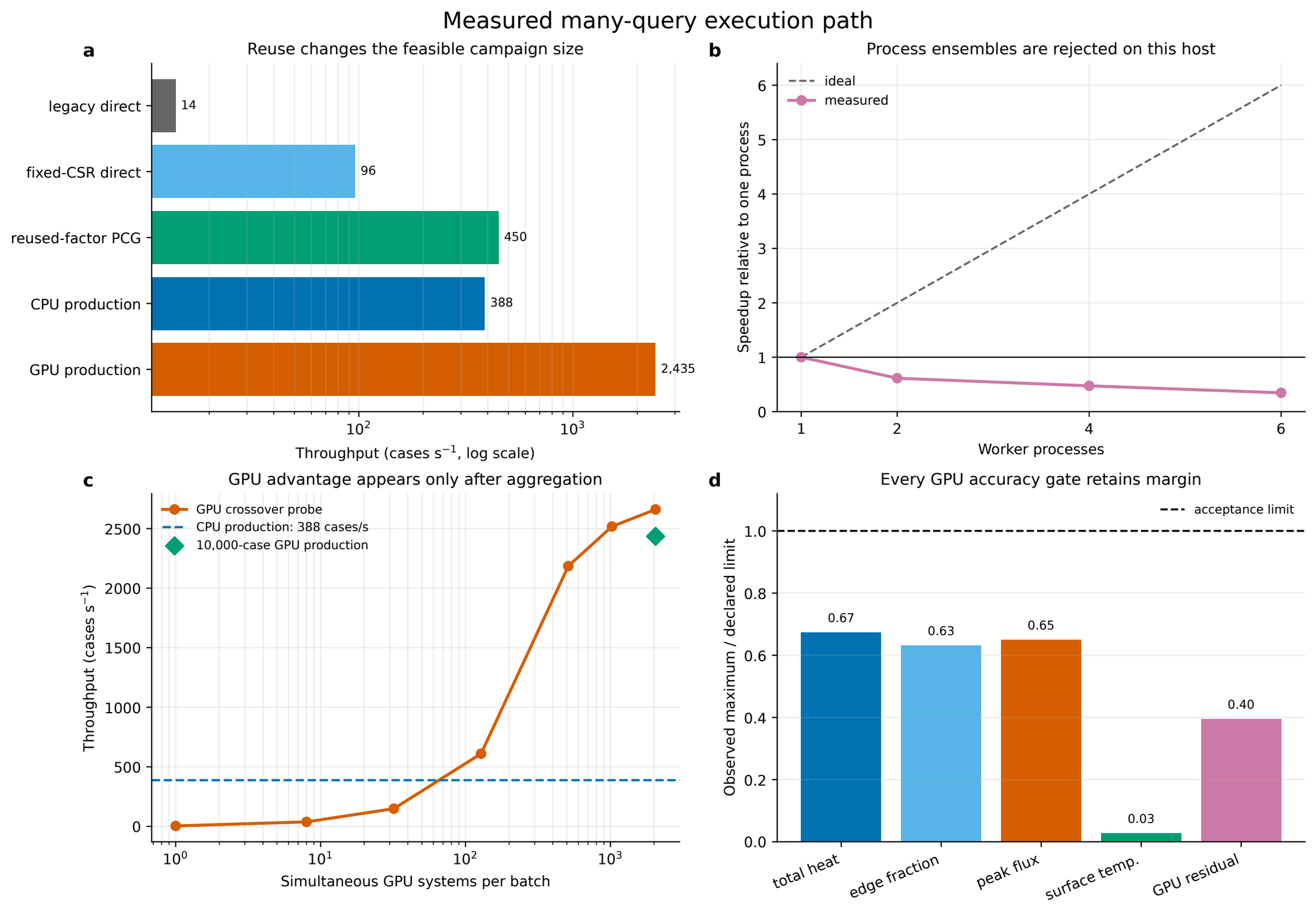
Measured many-query execution path
a
Reuse changes the feasible campaign size
legacy direct
fixed-CSR direct
reused-factor PCG
CPU production
GPU production
14
96
450
388
2,435
10^2
10^3
Throughput (cases s^{-1}, log scale)
b
Process ensembles are rejected on this host
ideal
measured
Speedup relative to one process
0
1
2
3
4
5
6
1
2
4
6
Worker processes
c
GPU advantage appears only after aggregation
GPU crossover probe
CPU production: 388 cases/s
10,000-case GPU production
Throughput (cases s^{-1})
0
500
1000
1500
2000
2500
10^0
10^1
10^2
10^3
Simultaneous GPU systems per batch
d
Every GPU accuracy gate retains margin
acceptance limit
Observed maximum / declared limit
0.0
0.2
0.4
0.6
0.8
1.0
0.67
0.63
0.65
0.03
0.40
total heat
edge fraction
peak flux
surface temp.
GPU residual